TECHNICAL ANNEX — EMISSION FACTORS PER DOLLAR OF SALES

Arden Environmental Quality Board, Air and Climate Division

1. Derivation

Factors are derived by dividing a physical emission factor for the product by a representative price. Physical factors are those in general use for stationary and mobile combustion and are reproduced in the reference table of the factor schedule. Representative prices are the volume-weighted average price of in-state sales of the product class over the basis year.

2. Products and factors

Three product classes are provided for: oil and refined products; natural gas; coal. The factor for coal is materially the highest per dollar of sales because coal is the least costly of the three per unit of energy and the most carbon-intensive. The factor for natural gas is the lowest of the three per dollar. There are no registered coal sales in Arden and the coal factor is provided for completeness.

3. Sensitivity to price

Because the denominator is a price, the factor moves inversely with price. Over the measuring period the volume weighted price of refined products in Arden varied by about 37.2 per cent peak to trough and of natural gas by about 39.7 per cent. The Division has adopted a single basis year rather than a period average, on the view that a single year is administrable and that the alternative invites dispute about the weighting.

4. Uncertainty

The Division's uncertainty estimate is 8 per cent for the refined products factor and 7 per cent for natural gas, at a one standard deviation level, arising principally from the price denominator rather than from the physical factor.

5. What the factors are not

The factors do not measure any entity's own emissions. They measure the emissions that would arise on combustion of the products the entity sold. Emissions from the entity's operations are not within the scope of the annex and are not part of the assessment as drafted.

6. Comparison with volume-based factors

The Division has computed, for the basis year, what the allocation would be on physical volume rather than on sales value, for the purpose of testing the price denominator. The two allocations differ by not more than 12.4 percentage points for any product class. The difference arises entirely from within-class price variation: a seller purchasing at the low end of the range and selling at the high end bears a smaller share on a value basis than on a volume basis. The Division does not regard the difference as large enough to justify collecting volume data that is not presently reported.

7. Products not separately provided for

The three classes do not exhaust the products within the statutory definition. Liquefied petroleum gases, residual fuel oil and petroleum coke are each combusted at factors materially different from the class average. The Division proposes to treat them within the oil and refined products class in the first instance, and to keep the treatment under review. Sales of these products in Arden are small.

8. What the factors assume

Three assumptions are built in and should be stated. First, that the product sold is combusted rather than used as a feedstock; a quantity used to make plastics is treated as though it were burned. Second, that combustion is complete. Third, that emissions from extraction, refining and transport are excluded; the factors are combustion factors only. Each assumption tends to overstate or understate in a direction the Division has not attempted to net out.

9. Revision

The Division proposes annual revision by reference to the volume-weighted price of in-state sales for the preceding year. A revision changes the allocation among covered entities even where no seller's conduct has changed. The Division has considered fixing the factors for the whole of the assessment period instead, and prefers annual revision on the ground that a fixed factor becomes progressively less related to physical quantity. Factor values are in the accompanying schedule. Revision is proposed annually.